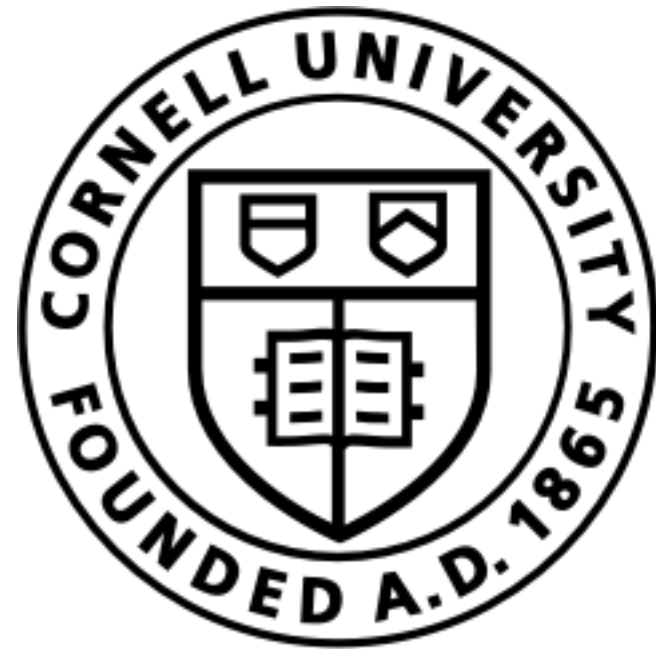




Lecture 5a: Expected Utility Hypothesis, Risk and Risk Aversion. The St. Petersburg Paradox and the Insurance Problem

CHEME 5760 Cornell University

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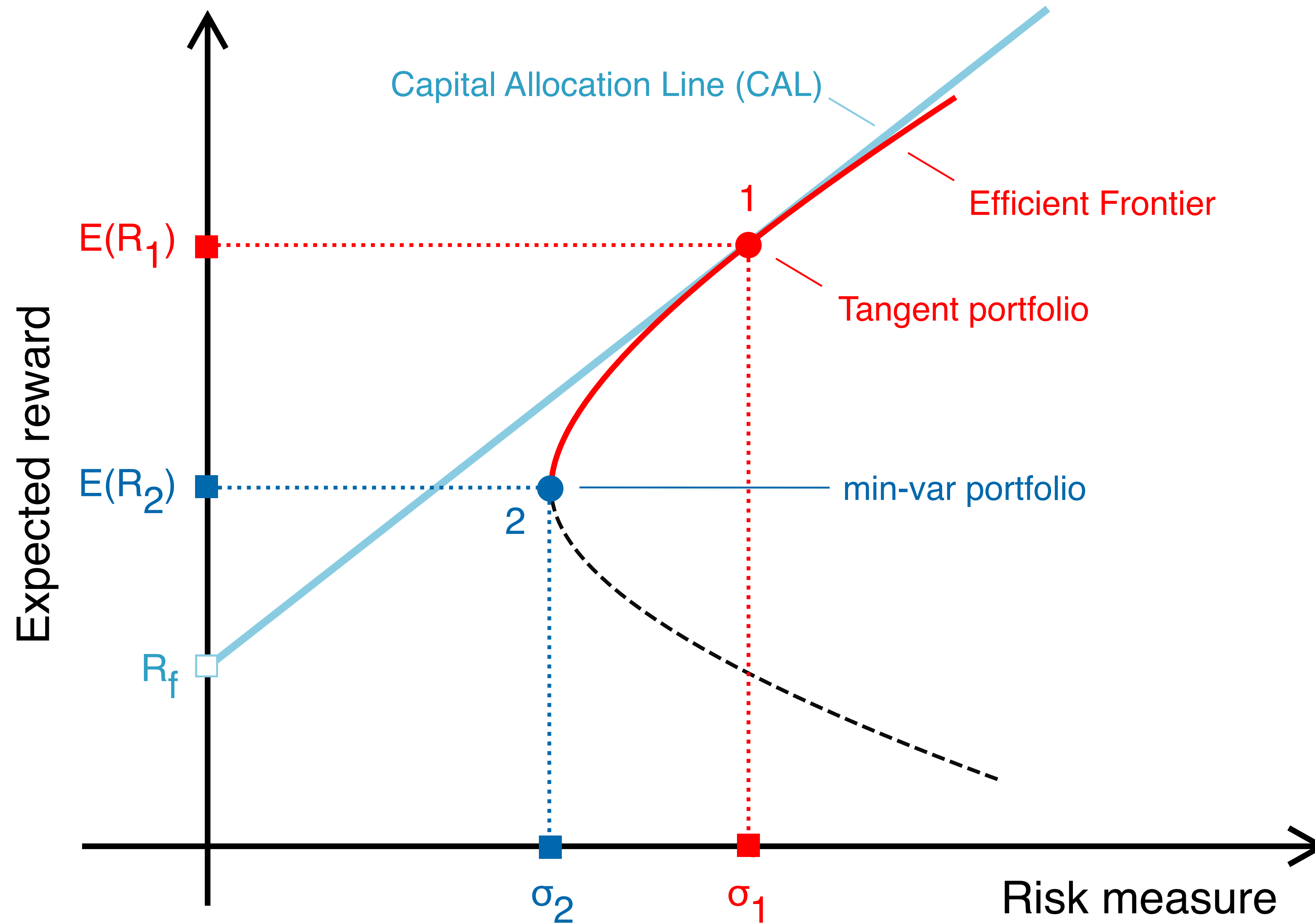
Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low-risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

The Markowitz allocation problem for a portfolio $\mathcal{P}$ composed of risky and risk-free assets minimizes the variance of the portfolio $\sigma_{\mathcal{P}}^2$ subject to a minimal required return and non-negativity constraints:

$$\begin{aligned} \text{minimize } \sigma_{\mathcal{P}}^2 &= \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{P}} w_i w_j \text{cov}(r_i, r_j) \\ \text{subject to } \mathbb{E}(r_{\mathcal{P}}) &= \left(w_f r_f + \sum_{i \in \mathcal{P}} w_i \cdot \mathbb{E}(r_i) \right) \geq R^* \\ \text{and } w_i &\geq 0 \quad \forall i \in \mathcal{P} \end{aligned}$$

The term $w_i \geq 0$ denotes the fraction of risky asset i in the portfolio $\mathcal{P}$, the quantity w_f denotes the weight of risk-free assets in the portfolio $\mathcal{P}$, r_f denotes the risk free rate or return, while R^* is the minimal required return for $\mathcal{P}$.



Objectives. In this lecture, we will:

- ▶ **O1:** The Markowitz portfolio problem is based on the expected return, i.e., the average possible return over many possible independent future events. *Is using expectation a good idea?*
- ▶ **O2:** Introduce the Insurance Problem, the Certainty Equivalent (CE), and the Risk Premium (RP). *How much would you be willing to pay for an insurance policy?*

————— next time —————

Revisit the risky + risk-free investment problem. Given your risk attitude, compute the optimal position on the capital allocation line.

Jupyter Notebooks

Repository: <https://github.com/varnerlab/CHEME-5760-Examples-F23.git>

- Example: CHEME-5760-L5a-StPetersburgParadox.ipynb
- Example: CHEME-5760-L5a-InsuranceProblem.ipynb

Monte-Carlo simulation of the St. Petersburg Paradox (Nicolas and Daniel Bernoulli, 1738)

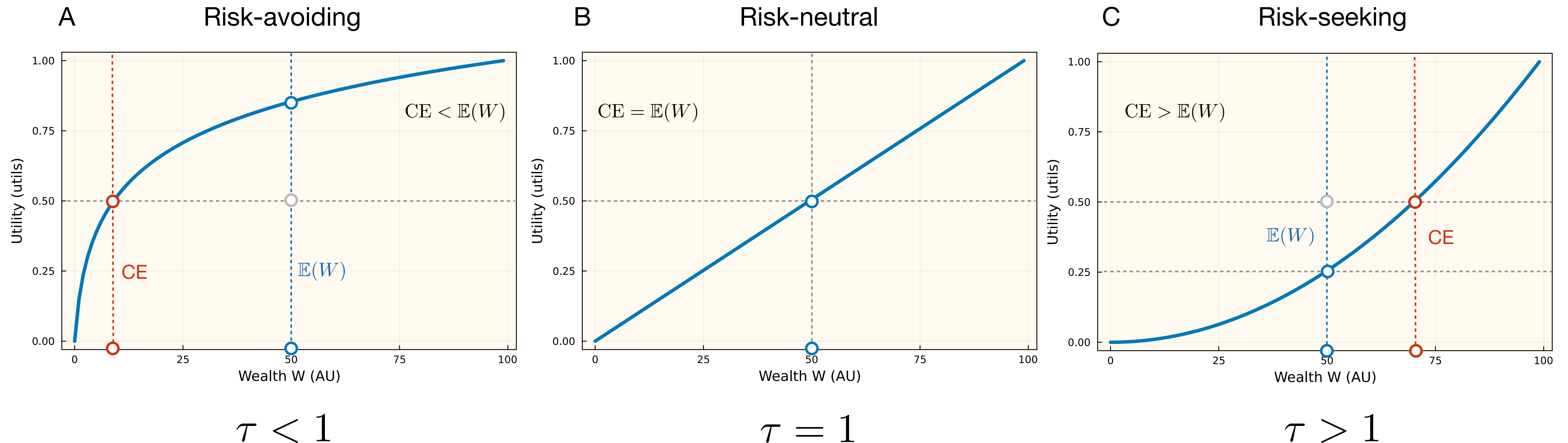
- ▶ A casino offers a single-player game in which a fair coin is tossed multiple times. The rules of the game:
 - The initial stake (pot) begins at \$1 and is doubled every time a head (H) appears.
 - The first time a tail (T) appears, the game ends, and the player wins the current stake.
- ▶ How much would you be willing to pay to enter this game?

Example: CHEME-5760-L5a-StPetersburgParadox.ipynb

O2: Introduce the Insurance Problem, the Certainty Equivalent (CE), and the Risk Premium (RP). **How much would you be willing to pay for an insurance policy?**

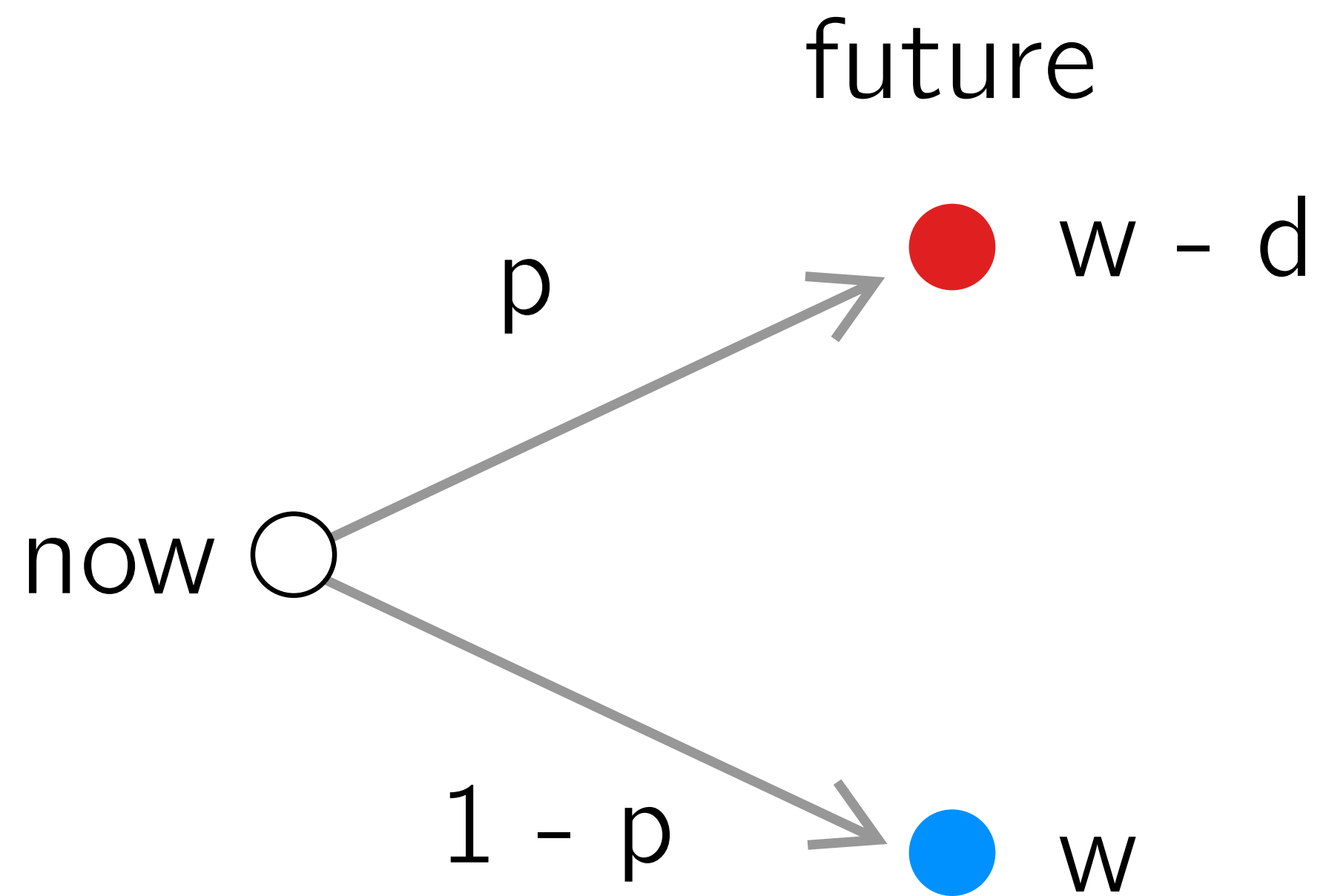
The **certainty equivalent (CE)** is a guaranteed level of wealth that provides the same utility as the expected utility of a risky scenario.

Consider: $U(w, \tau) \sim w^\tau$

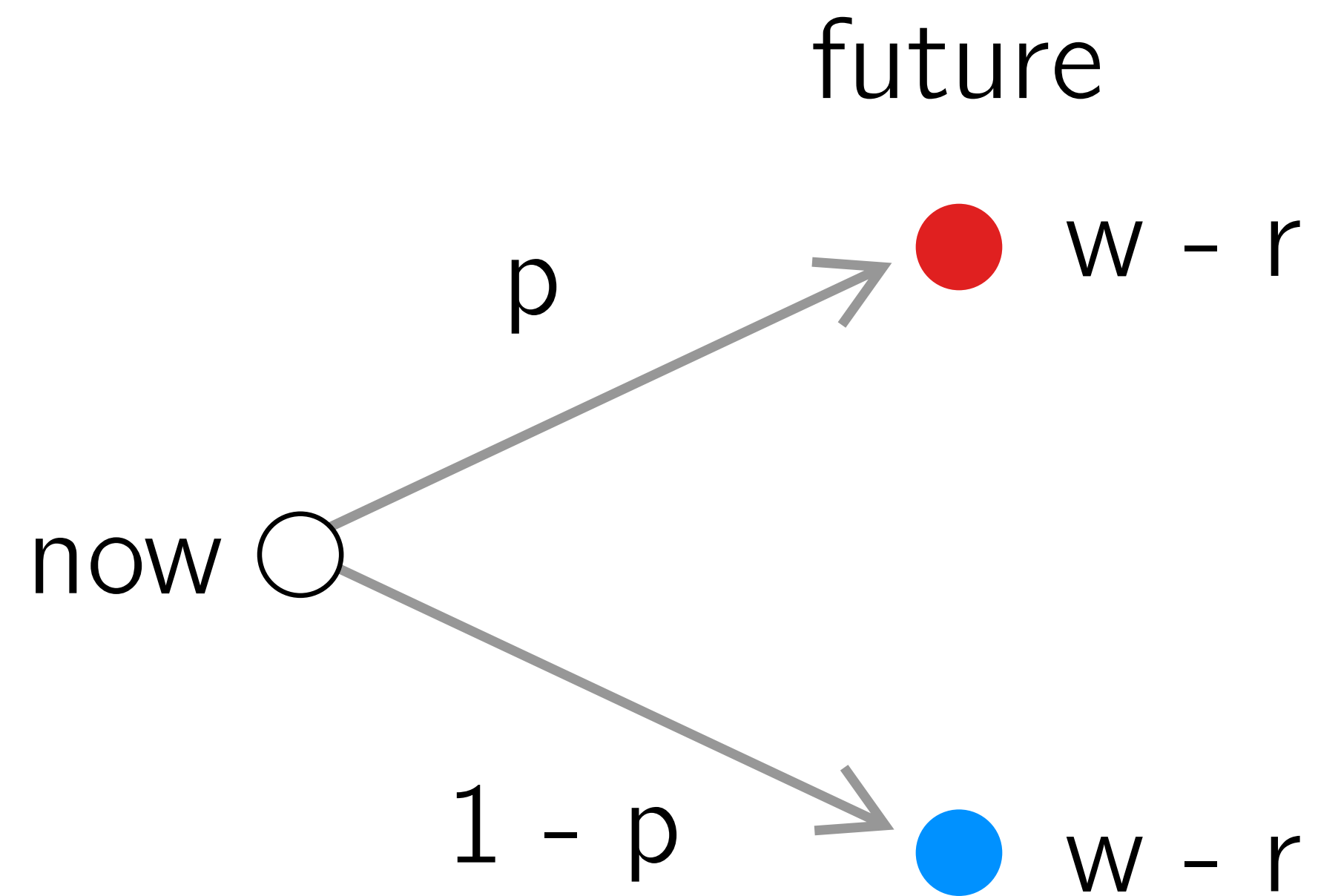


A decision-maker is indifferent between receiving the certainty equivalent and the risky scenario.

Concept: The Insurance Problem



A: without insurance



B: with insurance

Example: CHEME-5760-L5a-InsuranceProblem.ipynb

Today?